

Elementary Particle as a Closed Wave of Field 01

Normal Degree of Freedom, Mass, Memory, and the Absence of a Point Singularity

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Abstract

This paper formulates a first core element of Field 01: an elementary particle is interpreted not as a material point, but as a stable closed wave configuration of a field. Two regimes of the field—fixation 0 and direction 1—form a rhythm from which phase circulation, a scalar-profile / local-normal interpretation, mass, and memory can be described. A photon is interpreted as an open transport of phase 1, whereas a massive elementary particle appears only when the circulation closes and a local scalar/normal profile is retained. This approach replaces the point-like core by a dynamical core and thereby avoids an internal point singularity in the model description of an elementary particle. A black-hole horizon appears as a limiting regime of the same logic: suppression of the local scalar/normal profile and transition of phase memory into a surface record.

Keywords: Field 01; closed wave; elementary particle; phase circulation; scalar profile; normal-retention interpretation; mass; memory; quantum field theory; horizon; holography.

1 How to Read This Paper

This paper is a research text, not a completed physical theory. Its purpose is to make a first step from the book-level intuitive description of Field 01 toward a formal language that can be discussed, criticized, and gradually compared with established physics.

The reader should distinguish three levels of statements:

1. **Established physics:** quantum fields, vacuum correlations, photon energy $E = \hbar\omega$, mass–energy equivalence, the stress–energy tensor, and horizon entropy.
2. **Interpretation proposed by Field 01:** a particle is described as a stable field regime, a photon as open phase transport, and a massive particle as a closed node with a retained scalar/normal profile.
3. **Open hypothesis:** the scalar/normal profile interpretation, memory, phase tension, horizon surface recording, and the non-literal evaporation picture used in Field 01 require further mathematical formalization.

Therefore, all formulas not derived from a strict action principle or from a standard theory should be read as current notation and as directions for future formalization. Field 01 does not claim that the Standard Model is wrong, nor does it replace quantum field theory. It offers an additional phase-topological language whose consistency must be tested.

2 Aim of the Paper

Field 01 proposes to treat physical structures not as a collection of ready-made objects in a pre-given space, but as stable regimes of a single field. This paper isolates the minimal core of the model: a possible description of an elementary particle as a closed wave.

The main aim is to introduce a vocabulary that can connect the intuitive description with a mathematical framework. The proposed formalization is not presented as a replacement for the Standard Model. It is an interpretive and hypothetical layer that may help organize questions about particles, mass, memory, and horizons.

The central thesis is:

An elementary particle can be interpreted as a stable closed phase configuration of Field 01, in which the phase circulation is self-consistent and a local scalar/normal profile is retained.

3 Two Regimes of Field 01

Let $\mathcal{F}_{01}$ denote the field considered in the model. At the qualitative level it is described by two elementary regimes:

- 0 — fixation, holding, boundary, recording;
- 1 — direction, motion, circulation, phase transfer.

These symbols should not be read as binary digits in an ordinary computational sense. They are minimal names for two complementary modes of field behavior. The regime 1 expresses directional phase motion, while the regime 0 expresses the ability of the field to hold, close, or record a configuration.

A physical process in Field 01 is not the motion of an object through an empty background. It is a change of regime in the field itself. The basic rhythm can be written schematically as

$$0 \leftrightarrow 1. \tag{1}$$

This expression is not yet an equation of motion. It is a compact notation for the alternation between fixation and directed phase change.

4 Phase and Circulation

To describe a wave-like regime we introduce a phase variable

$$\varphi = \varphi(x, t), \tag{2}$$

where x denotes position in an effective space and t denotes an effective time parameter. At this stage both space and time are treated as descriptive variables, not as independently explained fundamental entities of the model.

A phase circulation along a contour $\mathcal{C}$ may be written as

$$\oint_{\mathcal{C}} d\varphi. \quad (3)$$

If the circulation does not close, the phase is transported without forming a localized node. If the circulation becomes self-consistent, a closed wave configuration can arise. A minimal topological condition for such a configuration is

$$\oint_{\mathcal{C}} d\varphi = 2\pi n, \quad n \in \mathbb{Z}. \quad (4)$$

This condition is not meant to define a complete particle spectrum. It only expresses the idea that a stable node requires a quantized or self-consistent phase return.

5 Photon as Open Circulation

In standard physics a photon is a quantum of the electromagnetic field, with energy

$$E = \hbar\omega. \quad (5)$$

In the language of Field 01 this relation can be read as a link between energy and the frequency of phase rhythm. A photon is interpreted as an open transport of the regime 1: phase is carried, but no local scalar/normal profile is retained.

Schematically,

$$\text{photon :} \quad \mathcal{N} = 0, \quad \text{open phase transport.} \quad (6)$$

This is not intended to replace quantum electrodynamics. It is a model-specific interpretation of why the photon has no rest mass: in Field 01, mass is associated with the retention of a closed phase structure and a local scalar/normal profile, while the photon remains an open propagating mode.

6 Elementary Particle as a Closed Wave

An elementary particle is interpreted as a stable closed configuration of the field. Unlike a photon, it is not merely a transfer of phase along an open direction. It is a self-returning phase regime.

The schematic condition for such a regime is

$$\oint_{\mathcal{C}} d\varphi = 2\pi n. \quad (7)$$

Here $\mathcal{C}$ does not have to be understood as a material wire or a literal circular path. It denotes the effective phase cycle that returns the field to a self-consistent state.

In this interpretation the particle is not a small ball and not a mathematical point with an infinite core. It is a localized mode of field rhythm. Its stability is produced by closure, not by a hard inner substance.

7 Normal Degree of Freedom and Local Depth

A closed phase circulation introduces an additional feature absent in the open photon mode: a local scalar/normal profile. In the current toy comparison, this should first be read as a scalar modulus / VEV-profile-like variable. We denote it by

$$\mathcal{N}. \quad (8)$$

This scalar/normal profile is not a material rod inside the particle. In Field 01 language it is a way of describing local depth, or the direction in which the closed phase configuration can hold itself as a localized node.

Qualitatively one may write

$$\mathcal{N} = \mathcal{N}[\Delta\varphi], \quad (9)$$

where $\Delta\varphi$ denotes the phase difference or phase gradient that supports the closed configuration.

Thus the distinction between a photon and a massive particle can be expressed as

$$\begin{aligned} \text{photon :} & \quad \mathcal{N} = 0, \\ \text{massive particle :} & \quad \mathcal{N} \neq 0. \end{aligned} \quad (10)$$

This is a current hypothesis of the model. A strict version would require a geometric definition of $\mathcal{N}$ and a dynamical equation for its stability.

8 Mass as Retention of the Normal

In standard physics mass is related to energy by

$$E = mc^2, \quad (11)$$

and, in the Standard Model, elementary particle masses are connected with their coupling to the Higgs field. Field 01 does not deny this structure. It asks whether mass can also be described in an additional phase-geometric language.

In the model, mass is interpreted as the energy cost of maintaining a closed phase configuration and its local scalar/normal profile. A schematic expression is

$$mc^2 \sim E_{\text{closed}}[\varphi, \mathcal{N}]. \quad (12)$$

A possible functional form is

$$E_{\text{closed}} = \int_{\Sigma} (A|\nabla\varphi|^2 + B|\mathcal{N}|^2 + V(\varphi, \mathcal{N})) d\Sigma, \quad (13)$$

where A , B , and V are not yet fixed from a fundamental action. The purpose of this expression is only to show the direction of formalization: mass should correspond to the energy of a stable closed solution, not to an independent substance placed inside the particle.

In words:

Mass is associated with the energy required to hold a closed phase rhythm and prevent it from unfolding into an open mode.

9 A Dynamical Core Instead of a Point Core

The usual problem of a point-like particle is that if all structure is concentrated at a mathematical point, singular behavior may appear in the description. Field 01 replaces the point core by a dynamical core.

The core of the particle is not a point at which quantities must diverge. It is the closed self-consistency of the phase circulation. The particle is localized not because it contains a hard center, but because its phase rhythm returns to itself.

This allows the model to avoid an internal point singularity at the level of interpretation. It does not by itself solve all ultraviolet problems of quantum field theory. Rather, it suggests a different geometric intuition: elementary localization should be described by stable closed field modes rather than by structureless points.

10 Memory as an Interphase Structure

Field 01 uses the term memory not as psychological memory and not as a hidden classical record inside a particle. Memory means the retained pattern of phase relations that makes a configuration identifiable and dynamically continuous.

Let

$$\mathcal{M} = \mathcal{M}[\varphi, \mathcal{N}, \mathcal{C}] \tag{14}$$

denote the information retained by a closed configuration. This notation is schematic. It means that memory is carried not by a material shape alone, but by phase relations, circulation, and the stability of the local scalar/normal profile.

This point is important for the later discussion of horizons. If memory is phase-structural rather than object-like, then a limiting surface can record the phase pattern without transporting a solid object through the horizon.

11 Forces as Regimes of a Closed Wave

The book-level formulation of Field 01 describes interactions in terms of how closed waves coordinate, hold, exchange, or open their phase structures. In this paper this idea is kept as a qualitative guide rather than as a completed derivation of the known interactions.

A cautious mapping is the following:

- electromagnetic interaction may be associated with phase asymmetry and open phase exchange;
- strong interaction may be associated with tight coordination of local scalar profiles / VEV-profile layers and local phase locking;
- weak interaction may be associated with instability, reconfiguration, or decay of a closed phase regime;

- gravity may be associated with collective deformation of the field and with the energetic cost of stable configurations.

This mapping is not a substitute for gauge theory. It is a research program: the next step would be to identify whether the phase variables of Field 01 can reproduce gauge symmetries, charges, and known conservation laws.

12 Relation to Established Physics

Quantum Field Theory

Quantum field theory already treats particles as excitations of fields. Field 01 is compatible with this broad idea, but proposes a specific interpretation: a massive particle is not merely any excitation, but a closed and self-consistent phase regime.

The model should therefore be compared with solitons, topological defects, bound states, and other stable field configurations. The analogy is useful, but not yet a derivation.

Vacuum Correlations

The vacuum in quantum field theory is not empty. It contains correlations and fluctuations. Field 01 uses this as a conceptual bridge: what appears as a particle may be treated as a stable pattern inside a deeper field structure.

No claim is made here that Field 01 has already derived the quantum vacuum. The claim is more modest: the language of phase structure and closure is not alien to modern field-based physics.

Spin

Spin is a quantum number with precise transformation properties under rotations and Lorentz symmetry. Field 01 does not yet derive spin. It suggests that spin may eventually be related to the topology of phase circulation and to the way a closed wave returns to itself.

A future formalization would have to show how half-integer and integer spin representations arise. Until then, the relation between spin and closed circulation remains a hypothesis.

Higgs Mechanism

The Standard Model explains elementary masses through coupling to the Higgs field. Field 01 does not replace this mechanism at the present stage. A possible bridge is to interpret mass as the energy of maintaining a stable closed configuration, while the Higgs mechanism supplies the established field-theoretic description of mass generation.

A serious comparison would require deriving effective mass terms from a Field 01 action and checking whether they can be related to known Higgs couplings.

Stress–Energy Tensor

In general relativity the stress–energy tensor describes the distribution of energy and momentum that sources spacetime curvature. Field 01 may eventually require its own effective stress–energy tensor for phase configurations.

At the current stage, however, expressions such as phase tension or scalar-profile retention energy are only provisional terms. They should not be confused with an already defined tensorial theory.

13 Limit: Vanishing Normal and Horizon

The same logic that distinguishes an open photon mode from a closed massive particle also suggests a limiting regime near a horizon. If the local scalar/normal profile is suppressed,

$$\mathcal{N} \rightarrow 0, \tag{15}$$

then the localized bulk-like structure can no longer be described as an ordinary closed particle in volume.

In Field 01 the horizon is interpreted as a recording surface. The closed configuration does not have to be imagined as mechanically torn apart. Rather, its phase memory can be transferred into a surface code:

$$\mathcal{M}_{\text{bulk}} \longrightarrow \mathcal{M}_{\text{boundary}}. \tag{16}$$

This is intentionally close to the general spirit of the holographic principle and to the Bekenstein–Hawking relation between entropy and horizon area. It is not presented as a derivation of black-hole thermodynamics. It is a model-specific interpretation of why information may be associated with a boundary rather than with an internal object-like volume.

14 Program for Further Formalization

To turn Field 01 into a mathematical theory, several tasks are necessary.

State Space

One must define the state space of the field:

$$\Psi_{01} \in \mathcal{H}_{01} \tag{17}$$

or an equivalent classical field configuration space. It must be clear what the fundamental variables are and how physical states are identified.

Action or Energy Functional

A dynamical principle is required. A possible schematic action is

$$S_{01} = \int \mathcal{L}(\varphi, \partial_\mu \varphi, \mathcal{N}, \partial_\mu \mathcal{N}) d^4x. \tag{18}$$

The Lagrangian must be chosen or derived so that stable closed configurations exist.

Topological Charge

Closed waves should be classified by a topological quantity, for example

$$Q = \frac{1}{2\pi} \oint_{\mathcal{C}} d\varphi. \quad (19)$$

A stable particle-like solution would correspond to a non-trivial value of Q .

Stability Condition

A particle should correspond to a stable or metastable extremum of the energy:

$$\delta E_{\text{closed}} = 0, \quad \delta^2 E_{\text{closed}} > 0 \quad (20)$$

within an appropriate class of variations.

Bulk-to-Boundary Transition

The transition toward a horizon should be described by a limiting map

$$\Pi_{\partial} : \mathcal{M}_{\text{bulk}} \rightarrow \mathcal{M}_{\text{boundary}}, \quad (21)$$

with the local scalar/normal profile suppressed:

$$\mathcal{N} \rightarrow 0. \quad (22)$$

Comparison with Known Quantum Numbers

Finally, the model must explain or reproduce known quantum numbers: charge, spin, mass, flavor, and gauge representation. Without this, Field 01 remains an interpretive framework rather than a predictive physical theory.

15 What Is Claimed and What Remains Hypothetical

The paper makes the following limited claims:

- it is possible to formulate a coherent phase-topological vocabulary for Field 01;
- an elementary particle can be interpreted as a closed phase configuration;
- the photon can be interpreted as open phase transport without a retained scalar/normal profile;
- mass can be associated with the energetic cost of maintaining a closed configuration;

- a horizon can be interpreted as a limiting surface where the local scalar/normal profile is suppressed and phase memory becomes boundary record.

The following points remain open:

- derivation from a precise action;
- reproduction of the Standard Model particle spectrum;
- derivation of spin and gauge charges;
- quantitative relation to the Higgs mechanism;
- construction of an effective stress–energy tensor;
- testable predictions that differ from existing theories.

Thus Field 01 should currently be read as a structured research program, not as a completed theory of nature.

16 Conclusion

The central idea of this paper is simple: an elementary particle may be described as a closed wave of Field 01 rather than as a point-like object. The closure of phase circulation produces a localized node, the retained scalar/normal profile gives the node local depth in the Field 01 interpretation, and the energy of maintaining this structure can be interpreted as mass.

A photon is then the limiting open case: it transports phase and energy, but does not retain a scalar/normal profile and therefore has no rest mass. A horizon is the opposite limiting case: the local scalar/normal profile is suppressed and the phase memory of a bulk configuration is transferred into a boundary record.

This language does not replace established physics. It provides a possible bridge between intuitive descriptions of particles, field-theoretic excitations, topological stability, and horizon information. The next task is to formulate the dynamics strictly enough that the model can be compared with known equations and, eventually, with observations.